Literature

Wiki *Enhanced dissipation for passive scalars*

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1 Orientation

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2 General mechanisms and rate limitations

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5. M. Coti Zelati and T. Gallay, *Enhanced dissipation and Taylor dispersion in higher-dimensional parallel shear flows*, Journal of the London Mathematical Society **108** (2023), 1358–1392, arXiv:2108.11192.

4 Circular and Hamiltonian flows and Taylor dispersion

1. M. Beck, O. Chaudhary and C. E. Wayne, *Analysis of enhanced diffusion in Taylor dispersion via a model problem*, Hamiltonian Partial Differential Equations and Applications, Fields Institute Communications **75** (2015), pp. 31–71, arXiv:1501.05877.

2. M. Beck, O. Chaudhary and C. E. Wayne, *Rigorous justification of Taylor dispersion via center manifolds and hypocoercivity*, Archive for Rational Mechanics and Analysis **235** (2020), 1105–1149, arXiv:1804.06916.

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4. M. Coti Zelati and M. Dolce, *Separation of time-scales in drift-diffusion equations on R^2*, Journal de Mathématiques Pures et Appliquées **142** (2020), 58–75, arXiv:1907.04012.

5. M. Coti Zelati, M. Dolce and C. Lo, *Diffusion enhancement and Taylor dispersion for rotationally symmetric flows in discs and pipes*, Journal of Mathematical Fluid Mechanics **26** (2024), Article 12, arXiv:2305.18162.

6. M. Dolce, C. J. P. Johansson and M. Sorella, *Dissipation enhancing properties for a class of Hamiltonian flows with closed streamlines*, Communications in Partial Differential Equations **50** (2025), 301–352, arXiv:2407.06884.

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8 Random stirring and transport noise

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